INCH-POUND

MIL-C-70991A (AR)
01 December 1993
SUPERSEDING
MIL-C-70991 (AR)
18 January 1991

MILITARY SPECIFICATION

CLOTH, COATED, KEVLAR-29, POLYURETHANE COATED

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations, and tests for polyurethane coated Kevlar-29 cloth. The coated cloth is used in the fabrication of the Ram Air Decelerator (RAD) of the M73/M75 grenade in the M261/M267 Hydra 70 submunition warhead (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-P-1136	-	Packaging and Packing of Coated (Plastic, Rubber) and Laminated Fabrics

MILITARY

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander U.S. ARMY ARDEC, ATTN.: SMCAR-BAC-S, Picatinny Arsenal, New Jersey, 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSMC N/A FSC 1330

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MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specifications for
MIL-W-63150	-	Weapons and Support Materiel Standard Quality Assurance Provisions for

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-595	-	Colors

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1487	-	Glossary of Cloth Coating Imperfections

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (SEE 6.5)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING

9345172	-	Fabric
9347191	-	Urethane
12938130	-	Coated Cloth

(Copies of Government documents, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

RULES AND REGULATIONS

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FEDERAL TRADE COMMISSION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Application for copies should be addressed to the Federal Trade Commission, Washington, D.C., 20580.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the document which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D2261	-	Tearing Strength of Woven Fabrics by the Tongue Method

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadephia, PA 19103-1187.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials.

3.1.1 Base cloth. The base cloth material shall meet all the requirements specified on drawing 9345172.

3.1.1.1 Yarn size. The base cloth shall have a yarn size of 200 denier.

3.1.1.2 Twist. The base cloth yarn shall have a 3.2 $\pm$ 0.5 Z twist in the warp and fill direction.

3.1.2 Urethane. The urethane shall meet all the requirements of drawing 9347191.

3.1.2.1 Coating composition. The coating composition shall be a thermosetting (crosslinked) polyether polyurethane coating or a cross-linkable base (tie) coat with a non-blocking PolyTetraMethyleneEtherGlycol (PTMEG) type top coat (see 6.6).

3.2. Physical Characteristics.

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3.2.1 Base cloth.

3.2.1.1 Weight. The base cloth shall weigh 1.93 to 2.31 oz/yd^2.

3.2.1.2 Thickness. The base cloth shall have a thickness of 0.007 inches maximum for the sample unit.

3.2.1.3 Breaking strength. The base cloth shall have a breaking strength of 275 pounds minimum in both the warp and fill direction.

3.2.1.4 Yarn count. The yarn count in the warp and fill direction shall be 40 $\pm$ 2 per inch.

3.2.1.5 Weave. The base cloth shall have a plain weave.

3.2.2 Coated cloth.

3.2.2.1 Coating weight. Coating shall be applied to both sides of fabric. Coating weight applied to side 1 (first side coated) shall be 1.0 $\pm$ .2 oz/yd^2.

3.2.2.2 Weight. The coated cloth shall weigh 3.4 to 4.1 oz/yd^2..

3.2.2.3 Thickness. The coated cloth shall be a maximum of 0.008 inches thick.

3.2.2.4 Breaking strength. The breaking strength of the coated cloth shall be a minimum of 275 pounds in both the warp and fill direction.

3.2.2.5 Tear strength. The tear strength of the coated cloth shall be a minimum of 35 pounds in both the warp and fill directions.

3.2.2.6 Burst strength. The coated cloth shall have a minimum burst strength of 650 pounds.

3.2.2.7 Cloth stiffness. The flex-stiffness of the coated cloth shall be 7.8 inches maximum in the warp direction and 6.4 inches maximum in the fill direction at 70° F and 8.9 inches maximum in the warp direction and 7.0 inches maximum in the fill direction at -55° F.

3.2.2.8 Hydrostatic resistance. The hydrostatic resistance of the coated cloth shall be a minimum of 480 pounds per square inch.

3.2.2.9 Blocking. The coated cloth shall exhibit little evidence of blocking; number of condition 3 not to exceed 27% and no condition 4, when subjected to the high temperature blocking test.

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3.2.2.10 Adhesion. The urethane coating shall adhere to the base cloth with a minimum of 9.8 pounds per inch.

3.2.2.11 Color. The color of the coated cloth shall be yellow. Numbers 13655, 23655 and 33655 of FED-STD-595 are representative of desired colors.

3.2.2.12 Length and put-up. Unless otherwise specified, the coated cloth shall be put-up on rolls as specified in PPP-P-1136. The length of the roll shall be as specified (see 6.2).

3.2.2.13 Width. The minimum width of the coated cloth shall be as specified (see 6.2).

3.2.2.14 Fabric identification. Each roll shall be labeled, tagged or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act as amended.

3.2.2.15 Workmanship. The finished coated cloth shall conform to the quality of product established by this specification and free of the defects as follows:

a. Any cut, hole, tear or any coated surface scratch or abrasion that results in baring the base fabric.

b. Any delamination of coating.

c. Foreign matter embedded in coating.

d. Crease or wrinkle resulting in doubling or adhesion of surfaces that cannot be corrected by manual pressure.

e. Coating cracked or checked.

f. Coating not applied over full width of base cloth (excluding selvage).

g. Any light area over 0.125 inch in diameter..

h. Any tackiness.

i. Any lump over 0.125 inch in diameter.

j. Any ripple, waviness, or dimensional distortion that cannot be corrected by manual pressure.

k. Any evidence of uncleanness.

3.3 First article inspection. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all the requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of any defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The lot to be produced for pulling first article sampling shall be a minimum of 100 linear yards. The first article sample shall be selected equally distributed from the beginning, center and end for coated cloth; and beginning and end for base cloth. Except for blocking tests, each sample unit shall consist of 5 specimens and the result shall be the average of the 5 specimens. For the blocking tests, each specimen shall be treated as a sample unit. Approximate quantities indicated below may be used for the performance of the first article test.

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Part Description	Quantity
Base Cloth	5 Square Yards
Coated Cloth	25 Square Yards

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components, and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification (see TABLE I) and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any sample unit fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any sample unit to comply with any of the requirements.

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70991A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Base Cloth	1 OF 3	9345172
			NEXT HIGHER ASSEMBLY: 12938130

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Materiel	COC	3.1.1	4.5.1
	Yarn size	COC	3.1.1.1	4.5.1
	Twist	COC	3.1.1.2	4.5.1
	Weight	6 (a)	3.2.1.1	4.5.2
	Thickness	6 (a)	3.2.1.2	4.5.3
	Breaking Strength	6 (a)	3.2.1.3	4.5.4
	Yarn count	6 (a)	4.2.1.4	4.5.5
	Weave	100%	3.2.1.5	Visual

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70991A(AR)

PARAGRAPH	TITLE		SHEET 2 OF 3	DRAWING NUMBER 12938130
	Coated Cloth			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Urethane material	COC	3.1.2	4.5.1
	Coating weight	6 (a)	3.2.2.1	4.5.2
	Weight	6 (a)	3.2.2.2	4.5.2
	Thickness	6 (a)	3.2.2.3	4.5.3
	Breaking Strength	6 (a)	3.2.2.4	4.5.4
	Tear strength	6 (a)	3.2.2.5	4.5.6
	Burst Strength	6 (a)	3.2.2.6	4.5.7
	Cloth stiffness	6 (a)	3.2.2.7	4.5.8
	Hydrostatic resistance	6 (a)	3.2.2.8	4.5.9
	Blocking	30 (b)	3.2.2.9	4.5.10
	Adhesion	6 (a)	3.2.2.10	4.5.11
	Color	100%	3.2.2.11	Visual
	Length	100%	3.2.2.12	Gage
	Width	100%	3.2.2.13	Gage
	Fabric Identification	100%	3.2.2.14	Visual

NOTES:

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70991A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Workmanship, Coated Cloth	3 OF 3	12938130
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Cut, hole, tear, or coated surface, scratch or abrasion	100% (c)	3.2.2.15a	Visual, light table MIL-STD-1487
	Delamination of coating	100% (c)	3.2.2.15b	Visual, MIL-STD-1487
	Foreign matter embedded in coating	100% (c)	3.2.2.15c	Visual, MIL-STD-1487
	Crease or wrinkle	100% (c)	3.2.2.15d	Visual, MIL-STD-1487
	Coating cracked or checked	100% (c)	3.2.2.15e	Visual, MIL-STD-1487
	Coating not applied over full width of base cloth (excluding selvage)	100% (c)	3.2.2.15f	Visual, MIL-STD-1487
	Light area	100% (c)	3.2.2.15g	Visual, MIL-STD-1487
	Tacky	100% (c)	3.2.2.15h	Visual, MIL-STD-1487
	Lump over 0.125 inches diameter	100% (c)	3.2.2.15i	Visual, MIL-STD-1487
	Ripple, waviness, or dimensional distortion	100% (c)	3.2.2.15j	Visual, MIL-STD-1487
	Evidence of uncleanness	100% (c)	3.2.2.15k	Visual, MIL-STD-1487

NOTES:

(a) Each sample unit shall consist of 5 specimens and the result shall be the average of the 5 specimens.
(b) The blocking test shall be performed on 15 individual specimens each, for both short term and for long term blocking test.
(c) Defects shall be defined in terms of linear yards of material. The smallest defect indicated shall be 1/8 (0.0125) linear yards. The percent defective for the sum of all workmanship defects shall not exceed 12% for the first article sample material. All defects shall be marked and not used.

AMSMC Form 1570a-E, 1 Jul 89 Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall consist of up to 1,000 linear yards of material, with overage allowable of 300 linear yards. A specimen is sufficient quantity of cloth required to conduct a particular test. The average of the required number of specimens, defined in FED-STD-191, shall constitute one sample. Unless otherwise specified this shall be 5 specimens. The result of a sample is the average of a required number of specimens. The required number of samples is defined in the Classification of Characteristics herein. Samples for the base cloth shall be equally taken from the beginning and end of the inspection lot or at the 500 linear yard intervals for a minimum or two (2) samples per inspection lot. Samples for first side coating weight shall be taken from the beginning and end of the inspection lot. Samples for coated cloth shall be equally taken from the beginning, center and end of the inspection lot.

4.4.2 Examinations and tests.

a. Classification of Characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality assurance program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing, as a minimum, the conformance criteria specified.

b. Conformance criteria. The accept/reject criteria corresponding to the number of samples (called out under the conformance criteria) in the Classification of Characteristics paragraphs shall be accept on zero and reject on one defect for all examinations and tests except were notes apply.

c. Alternative inspection provisions. Unless otherwise specified herein, or provided for in the contract, alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.7). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In case of disputes as to whether the contractor's proposed alternatives(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or inspection system, as applicable.

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-70991A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 9345172
4.4.2.1	Base Cloth		NEXT HIGHER ASSEMBLY 12938130

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Materiel	COC	3.1.1	4.5.1
102	Yarn size	COC	3.1.1.1	4.5.1
103	Twist	COC	3.1.1.2	4.5.1
104	Weight	2 (a)	3.2.1.1	4.5.2
105	Thickness	2 (a)	3.2.1.2	4.5.3
106	Breaking Strength	2 (a)	3.2.1.3	4.5.4
107	Yarn count	2 (a)	4.2.1.4	4.5.5
108	Weave	2 (a)	3.2.1.5	Visual
Minor	None defined.			

NOTES:

(a) Each sample unit shall consist of 5 specimens and the result shall be the average of the 5 specimens.

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-70991A(AR)

PARAGRAPH	TITLE		DRAWING NUMBER 12938130
4.4.2.2	Coated Cloth	SHEET 1 OF 1	NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Urethane materiel	COC	3.1.2	4.5.1
102	Coating weight	3 (a)	3.2.2.1	4.5.2
103	Weight	3 (a)	3.2.2.2	4.5.2
104	Thickness	3 (a)	3.2.2.3	4.5.3
105	Breaking Strength	3 (a)	3.2.2.4	4.5.4
106	Tear strength	3 (a)	3.2.2.5	4.5.6
107	Burst Strength	3 (a)	3.2.2.6	4.5.7
108	Cloth stiffness	3 (a)	3.2.2.7	4.5.8
109	Hydrostatic resistance	3 (a)	3.2.2.8	4.5.9
110	Blocking	15 (b)	3.2.2.9	4.5.10
111	Adhesion	3 (a)	3.2.2.10	4.5.11
Minor				
201	Color	3 (a)	3.2.2.11	Visual
202	Length	Each roll	3.2.2.12	Gage
203	Width	Each roll	3.2.2.13	Gage
204	Fabric Identification	Each roll	3.2.2.14	Visual

NOTES:

(a) Each sample unit shall consist of 5 specimens and the result shall be the average of the 5 specimens.
(b) The blocking test shall be performed on 15 individual specimens each, for short term only.

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-70991A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 12938130
4.4.2.3	Workmanship, Coated Cloth		NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Cut, hole, tear, or coated surface, scratch or abrasion	100% (a)	3.2.2.15a	Visual, light table MIL-STD-1487
102	Delamination of coating	100% (a)	3.2.2.15b	Visual, MIL-STD-1487
103	Foreign matter embedded in coating	100% (a)	3.2.2.15c	Visual, MIL-STD-1487
104	Crease or wrinkle	100% (a)	3.2.2.15d	Visual, MIL-STD-1487
105	Coating cracked or checked	100% (a)	3.2.2.15e	Visual, MIL-STD-1487
106	Coating not applied over full width of base cloth (excluding selvage)	100% (a)	3.2.2.15f	Visual, MIL-STD-1487
107	Light area	100% (a)	3.2.2.15g	Visual, MIL-STD-1487
108	Tacky	100% (a)	3.2.2.15h	Visual, MIL-STD-1487
Minor				
201	Lump over 0.125 inches diameter	100% (a)	3.2.2.15i	Visual, MIL-STD-1487
202	Ripple, waviness, or dimensional distortion	100% (a)	3.2.2.15j	Visual, MIL-STD-1487
203	Evidence of uncleanness	100% (a)	3.2.2.15k	Visual, MIL-STD-1487

NOTES:
(a) Defects shall be defined in terms of linear yards of material. The smallest defect indicated shall be 1/8 (0.0125) linear yards. The percent defective for the sum of all workmanship defects shall not exceed 12% for the first article sample material. All defects shall be marked and not used.

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4.4.3 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.4.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attribute data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078 and 6.4 herein.

4.5 Methods of inspection.

4.5.1 Materials. Provide a Certificate of Conformance (COC) as defined in MIL-W-63150 for all materials specified in 3.1.

4.5.2 Cloth weight/coating weight. This test shall be conducted in accordance with FED-STD-191 method number 5041. The ounces per square yard method shall be used in the calculation of results. Coating weight shall be determined by subtracting the average cloth weight from the individual coated cloth weight value.

4.5.3 Thickness. This test shall be conducted in accordance with FED-STD-191 method number 5030.

4.5.4 Breaking strength. This test shall be conducted in accordance with FED-STD-191 method number 5100, with the following exception. The test equipment shall be operated at a uniform pulling speed of 2 + 0.5 inches per minute.

4.5.5 Yarn count. This test shall be conducted in accordance with FED-STD-191 method number 5050.

4.5.6 Tear strength. This test shall be in accordance with ASTM D-2261. For a test to be valid, failure of the fabric fibers shall originate at the end of the pre-cut slit. Failures that originate at the clamp attachment or any other location other than the end of the pre-cut slit shall be termed a no-test and the test repeated on another sample specimen.

4.5.7 Burst strength. This test shall be conducted in accordance with FED-STD-191 method number 5122 with the following exception. A rubber gasket 1/16 inch thick with a one (1) inch diameter hole shall be used between the test specimen and the upper clamp of the testing machine with the holes in the rubber gasket and the upper ring clamp being visually concentric. Test specimens where the burst initiates against the edge of the opening in the upper ring clamp shall be classified as a no test.

4.5.8 Cloth stiffness. This test shall be conducted in accordance with FED-STD-191 method number 5204.

4.5.9 Hydrostatic resistance. This test shall be conducted in accordance with FED-STD-191 method number 5512 with the following exception. The o-ring in the test apparatus shall have a cross section of .139 $\pm$ .004 inch. A rubber gasket 1/16 inch thick with a one (1) inch diameter hole shall be used between the test specimen and the upper clamp of the testing machine

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with the holes in the rubber gasket and the upper ring clamp being visually concentric. Test specimens where the burst initiates against the edge of the opening in the upper ring clamp shall be classified as a no test.

4.5.10 Blocking. This test shall be conducted in accordance with FED-STD-191 method number 5872, with the following exceptions. The specimen shall be a square of cloth 2 by 2 inches. In addition the specimen shall be clean and free of dust. Two plates of smooth brass, or polished steel approximately 1.25 by 1.25 by 0.125 inches, or glass plates approximately 1.25 by 1.25 by 0.25 inches shall be used. A twenty pound weight shall be used. The specimen shall be placed in the oven for 6 hours at a temperature of $212^{o} \pm 5^{o}$ F.

4.5.10.1 Long term blocking. This test shall be conducted in accordance with 4.5.10 with the following exception. The specimen shall be dusted with talcum powder on both sides prior to folding and then placed in the oven for at least 7 days duration. This test is to be performed for first article only.

4.5.11 Adhesion. This test shall be conducted in accordance with FED-STD-191 method number 5970 with the following exception. The pulling speed shall be 2 inches $\pm$ 0.5 inches per minute.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2 and 6.2.1).

5.1.1 Levels A and C. The cloth, put-up as specified (see 3.2.2.12), shall be packaged in accordance with the applicable requirements of PPP-P-1136.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A, B or C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1136.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1136.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cloth is intended for use in the manufacture of inflatable retarding devices, Ram Air Decelerator (RAD), in munitions.

6.2 Acquisition requirements. Acquisition documents must specify the following:

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a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for submission of first article sample.

d. Width requirements.

e. Selection of applicable levels of packaging and packing

f. Length of roll

g. Quantity in yards

6.2.1 Packaging options. In the preparation of contractors or orders, it must be noted that various options, choices and alternatives as indicated in PPP-P-1136 may be exercised in the preparation for delivery of the cloth.

6.3 Consideration of Data requirements. The following data requirements should be considered when this specification is applied on a contract, per the Contract Data Requirements List (CDRL), DD Form 1423.

Reference paragraph	DID Number	DID Title	Suggested Tailoring
4.5.17	OT-90-12016	Inspection and Test Report	---

6.4 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN.: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List (CDRL), DD Form 1423 in the contract.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, US Army Armament Research and Development Command (ARRADCOM) or US Army Armament Research and Development Center (ARDC) drawings. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.6 Coating composition. The thermosetting (crosslinked) polyether polyurethane coatings are required in order to minimize coating adhesion failures and long term storage problems.

6.7 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

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6.8 Subject term (key words) listing.

Hydra 70 Rocket
RAD
M261 Submunition Warhead
M267 Submunition Warhead
M73 Grenade
M75 Grenade
Drag Device

6.9 Changes from the previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1330-A381)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-70991A (AR)	931201

3. DOCUMENT TITLE: CLOTH, COATED, KELVAR-29, POLYURETHANE COATED

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	e. DATE SUBMITTED (YYMMDD)

7. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial: 201-724-6675 (2) AUTOVON: DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 — *Previous editions are obsolete.* — 198/290

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